From SEED to SPRING

An Instructional Module for Middle School Classrooms

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1. Introduction

The instructional module "From SEED to SPRING" is designed to support middle school educators in implementing a classroom-based agricultural data science project, where SEED stands for Science, Environment, Exploration, and Data. Developed as a part of the IoT4AG (Internet of Things for Agriculture) initiative, this instructional unit guides students through a structured 10-day sequence that integrates plant growth, sensor technology, and environmental data analysis. The core aim is to provide students with hands-on, inquiry-based learning opportunities that align with state standards and promote real-world STEM connections.



[1]

While the module is designed for a 10-day implementation, it is intentionally flexible. Teachers may extend or adapt the timeline based on their instructional goals, classroom schedule, and available resources.

The module includes the following core components:

1. Introduction:
- SPRING Project Overview
    - An introduction to the SPRING project and its educational purpose.
- Technical Tools
    - A brief explanation of the key tools used in the module – such as environmental sensors (e.g., soil moisture), LoRaWAN gateway, and the dashboard – designed to make data collection and analysis accessible for classroom use.
2. Guideline:
- Teacher Guideline
    - Detailed guidance for setting up the in-class experiment, including recommended materials, suggested equipment, and step-by-step instructional support. Video tutorials are included to assist with preparation and delivery.



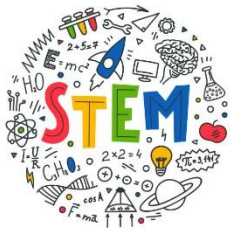
[2]

3. Implementation:
- Students' Engagement and Learning Material
    - Outlines how students engage with the experiment across the 10-day period. Lesson plans are organized using 5E instructional model and include guided activities, data analysis tasks, and reflection opportunities that encourage critical thinking, collaboration, and STEM skill development.

1.1. SPRING Overview

The Solar-Powered Remote IoT4Ag Network Gateway (SPRING) is originally developed as part of the IoT4Ag – an NFS-funded research initiative led by Purdue University in collaboration with partner institutions. The goal of the original SPRING system is to support agricultural decision-making by enabling real-time data collection from sensors across farms, even in rural areas with limited internet connectivity. Using solar-powered gateways and LoRaWAN (Long Range Wide Area Network) communication, SPRING helps researchers and producers access environmental data (such as soil moisture and temperature) through cloud-based dashboards.

[3]



[4]

This classroom module – From SEED to SPRING – adapts the principles and structure of the original SPRING system for middle school learning environments. While maintaining key technological components such as sensors, wireless transmission, and dashboard-based visualization, the educational version is designed for hands-on, student-centered inquiry within a 10-day instructional sequence.

In the classroom setting, the SPRING Project will be:

- A scaled-down model of agricultural technology made accessible and meaningful to students.
- A tool to introduce core STEM concepts such as environmental sensing, data interpretation, and critical thinking.
- A bridge between classroom learning and real-world innovation, showing students how science and technology contribute to sustainability.

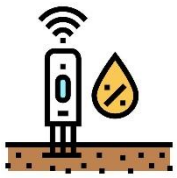
By using this adapted version of SPRING, educators can engage students in meaningful scientific exploration while building soft skills like digital literacy, collaboration skills, and environmental awareness - all aligned with academic standards.

1.2. Technical Tools



The classroom adaptation of SPRING incorporates simplified, accessible technologies that reflect the core principles of real-world agricultural systems. These tools enable students to explore environmental sensing, data collection, and basic data interpretation through hands-on experimentation and observation.

Environmental Sensors



[5]

At the heart of the classroom setup are environmental sensors—primarily soil moisture sensors—that measure water levels in the soil. These sensors allow students to monitor changing conditions, analyze trends, and understand how soil moisture affects plant growth. While the module is designed around soil moisture as the primary variable, teachers may choose to incorporate additional sensors, such as temperature, humidity, or light, to extend the inquiry depending on instructional goals and available resources.

Gateway and Data Transmission



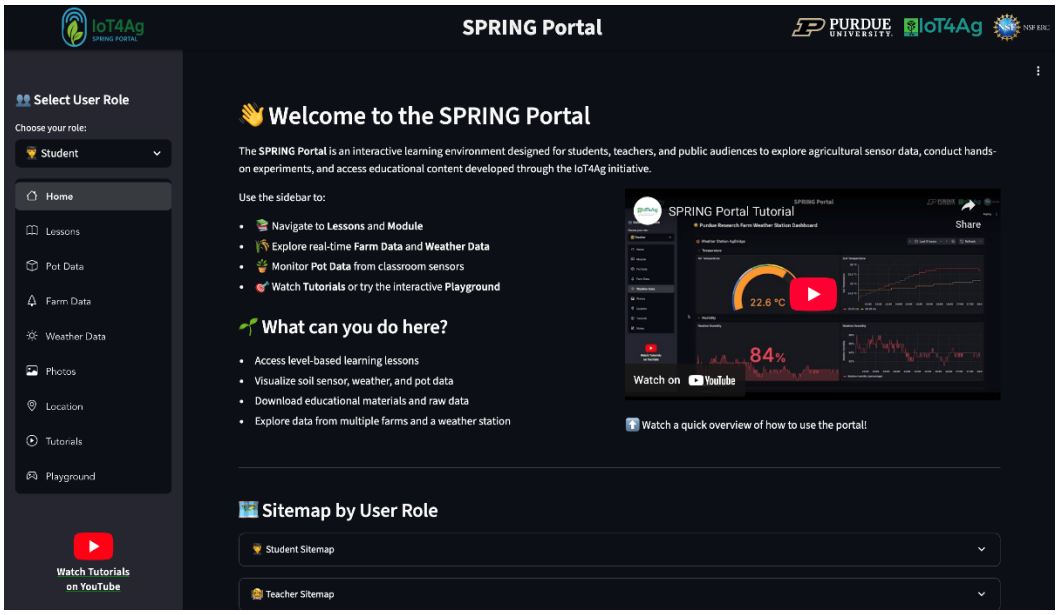
In the original SPRING system, LoRaWAN (Long Range Wide Area Network) is used for transmitting sensor data over long distances across agricultural fields (Bailey et al., 2024). In the classroom model, this function is adapted using the raspberry pi RAK gateway.

This gateway acts as a hub that collects data from connected sensors and transmits it to a cloud-based dashboard. This streamlined system allows students and teachers to access real-time data on their devices, visualize environmental changes, and engage with digital tools in meaningful ways.

Dashboard

IoT4Ag SPRING PORTAL

The dashboard (project website) allows teachers and students to view real-time sensor data, such as soil moisture levels, through an interactive interface. Users can track changes over time and visualize trends using simple graphs and charts.



2. Teacher Guideline



[6]

This section provides detailed step-by-step instructions to assist educators in preparing and managing the hands-on classroom experiment. Each stage of the setup process is supported by short video tutorials, accessible via the provided links. Teachers are encouraged to review both the written and visual materials in advance to ensure smooth and confident facilitation of classroom activities.

2.1. Materials and Equipment Preparation

1. Selecting the Pot and Soil

A pot with a diameter of approximately 16 inches is recommended, as it offers adequate space for both root development and sensor placement. Functional drainage holes are essential; if they are not pre-existing, they may be gently added. For soil, a basic top-soil variety without added fertilizers or nutrients is preferred, allowing for consistent control of experimental variables.



[7]

2. Choosing Seeds



Seed selection may be guided by student interests or local community partnerships. Sourcing seeds locally can enhance student engagement and foster broader outreach opportunities. In this module, corn serves as the example crop. The corresponding video tutorial demonstrates how to determine appropriate planting depth and achieve even water distribution across the soil surface.

2.2. Technology Setup

3. Gateway Configuration

The gateway functions as a communication bridge between the sensor and the data dashboard. Setup involves connecting the antenna, supplying power, and linking the device to the internet via Ethernet cable. Once operational, indicator lights confirm successful startup and connectivity.



[8]



4. Sensor Activation

To prepare the soil moisture sensor, the radio antenna is first attached. Holding the pairing button until a blinking green light appears indicates that the sensor has successfully connected to the gateway and is ready to transmit data.

2.3. Launching the Experiment

5. Soil Preparation and Planting Process

Prior to planting, the pot is filled with approximately three-quarters full of soil. Planting holes are then measured and created based on the crop's requirements. Seeds are placed, gently covered, and the soil is watered evenly to initiate germination. The video tutorial provides a visual reference for this process. While the use of gloves is optional, it may be helpful for comfort and hygiene.



[9]

6. Sensor Placement and Final Adjustments



The sensor should be positioned at root level, typically within the upper quarter of the soil. Ensuring the sensor remains stable throughout the experiment is essential for accurate data collection. Once this step is complete, the setup is ready for ongoing observation and student engagement.

Note: Full setup tutorials are available on our YouTube channel: youtube.com/@iot4agspring

2.3. Adapting the SPRING on Available Resources

This module was designed with flexibility in mind to ensure all classrooms — regardless of equipment — can meaningfully engage with the core content. Depending on what materials are available, teachers can adapt the implementation as follows:

- **If you have access to the full SPRING system (sensor + gateway):**

You can set up real-time data collection in your classroom. If needed, contact your school's administration or tech coordinator to request support in obtaining these tools.



[10]

- **If you don't have access to the gateway and sensor:**

You can still run a hands-on experiment using simple classroom moisture meters. Students can collect data manually, compare it to Purdue farm data via the dashboard, and continue following the activities with the same learning goals.

- **If no physical setup is possible:**

The module can still be implemented using dashboard data only. Several lessons are designed to work entirely with existing data from Purdue's SPRING experiment, so students can engage in observation, analysis, and solution design without planting in class.

There's always a way to bring innovation into your classroom — whether high-tech or low-tech. We believe in your creativity and your students' potential.

Note: Please take a picture of your plant every day, so you can track its growth and use the photos in future lessons.

3. Standard Covered in the Module

3.1. Agriculture Science and Business Standards

Domain- Plant, Soil, and Horticulture Science

Core Standard 3 Students have an understanding of plant and soil science and how it relates to the agricultural industry.

Standards

EA-3.1 Understand basic needs of plant growth from germination to soil fertility EA-3.4 Identify basic soil properties

Domain- Power, Structure, and Technical Systems

Core Standard 8 Students gain a basic understanding of agricultural power, structure, and technical systems.

Standards

EA-8.3 Explore modern technology applications within the agriculture industry

Domain – Careers

Core Standard 9 Students examine the scope of career opportunities in and the importance of agriculture to the economy.

Standards

EA-9.2 Describe career opportunities and means to achieve those opportunities in agriculture [11]

4. Lesson Plans



[12]

This module provides two coordinated sets of lesson plans: one focused on agricultural learning (10 days). The lessons are designed for classroom implementation using real-world data collected from plant sensors and a live digital dashboard.

Each lesson follows the 5E instructional model, a student-centered approach that supports exploration, concept-building, and reflection. Lessons are aligned with Indiana Academic Standards and integrate core vocabulary, daily objectives, and hands-on activities using classroom experiments and agricultural data.

4.1. What is the 5E Instructional Model?

The 5E model organizes instruction into five phases that reflect how students naturally learn:

- Engage: Capture students' interest and activate prior knowledge.
- Explore: Allow students to investigate a concept through hands-on or inquiry-based activities.
- Explain: Guide students to formalize their understanding with support from the teacher.
- Elaborate: Extend learning through new challenges, real-world connections, or applications.
- Evaluate: Assess students' understanding through tasks, discussions, or reflections.

This structure helps ensure that lessons are both meaningful and developmentally appropriate. Students build understanding through observation, discussion, experimentation, and data analysis- while teachers support and scaffold learning across each phase.

SEED to SPRING Lesson 1

Subject: Middle School Agriculture	**Standards:** EA-3.1 – Understand basic needs of plant growth from germination to soil fertility
Topic: What Do Plants Need to Grow?	**Lesson Objective:** Students will identify and describe the basic needs of plants (light, water, air, nutrients, space, and temperature) and explain how they support plant growth.
Materials: Design Your Own Plant Activity Sheet What If Game Shout The Answer Game	**Vocabulary:** Light: Energy from the sun that helps plants make food Water: Helps plants stay alive and move nutrients Air: Plants use carbon dioxide from air to grow Temperature: Affects how fast or slow plants grow Space: Room for roots and leaves to grow Nutrient: Substances in soil that help plants grow Plant health: How strong and green a plant is Growth: The process of a plant getting bigger

Engage

Begin by asking:

- Have you ever planted something before? Did it grow? Why or why not?1
- What do you think a plant needs in order to grow?

Allow a few students to share ideas. Write key words on the board (you'll return to these later).

Explore

Distribute the "Design Your Own Plant" activity sheet (Page 1).

Students will:

- Draw their own version of a healthy plant in a pot
- Use clues from the word bank to think about what the plant needs
- Share and discuss their drawings with a partner or group

Ask a few students to present their plant drawings and explain what they included and why.

Explain

Go over each vocabulary word together.

- Read the definition aloud and write it on the board.
- As each word is explained, have students point to or label that part on their plant drawing (Page 1).
- Then, students fill in the matching definitions on Page 2 of their activity sheet.

Elaborate

Play the "What If?" acting game:

- Divide students into small groups.
- Groups create and perform a quick skit showing what might happen.
- The class guesses which plant need was missing or unbalanced.

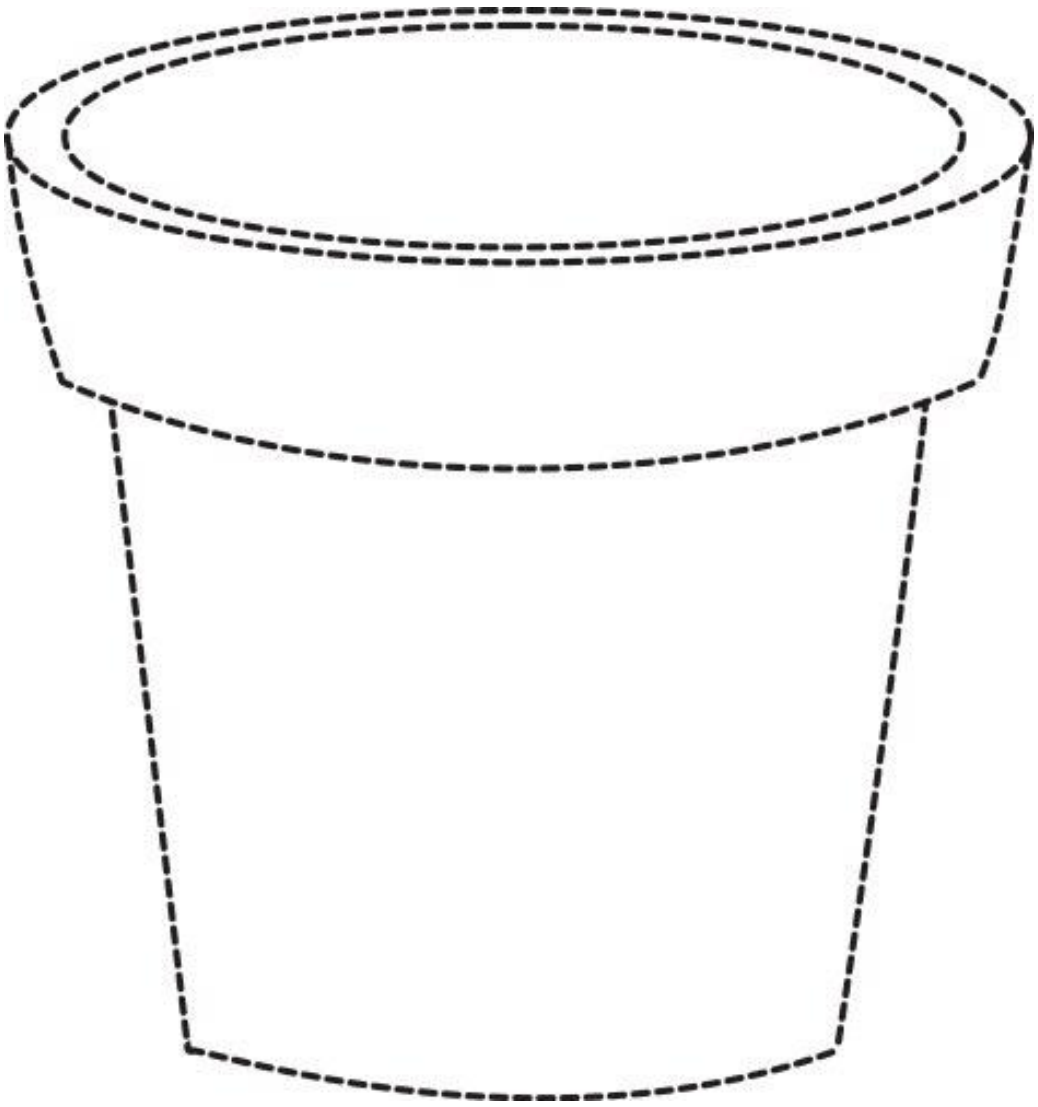
Evaluate

Play the "Shout the Answer" game:

- The teacher reads a short definition (e.g., "This helps plants make food using sunlight!")
- Students shout the correct vocabulary word as a group.

Notes

Design Your Own Plant



Light:……

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Water:……..……

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Air:……..……

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Temperature:……..……

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Space:……..……

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Nutrient:……..……

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Plant_health:……

……..……

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Growth:……..……

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What If Game

Purpose: Reinforce understanding of plant needs by imagining what happens when one is missing or unbalanced.

Instructions:

- Divide students into small groups (3–4 students per group).
- Give each group a scenario card or verbal prompt (e.g., "What if the plant didn't get enough water?" or "What if it had no space to grow?").
- Each group will quickly plan and act out a 15–30 second skit to show what happens to a plant in that situation.
- The rest of the class will try to guess which plant's need is missing or unbalanced.

After each performance, ask:

- What need was missing?
- How did it affect the plant?

Example Scenarios:

- Too much water
- No nutrients
- No air
- No sunlight
- No space to grow
- Temperature too cold

Shout The Answer Game

Purpose: Quick and fun whole-class review of vocabulary.

Instructions:

- Teacher reads aloud a short definition or clue from today's vocabulary.
- Students shout the correct word all together (as a class).

Example Prompts:

- This is in the soil and acts like vitamins for plants! → Nutrient

Tips:

- Encourage full class participation.
- You can turn it into a team challenge by keeping score if time allows.

SEED to SPRING Lesson 2

Subject: Middle School Agriculture

Standards:
EA-3.1 – Understand basic needs of plant growth from germination to soil fertility
EA-8.3 Explore modern technology applications within the agriculture industry

Topic: Seed Germination & Planting

Lesson Objective:
Students will describe the process of germination, plant their own seeds, and begin monitoring growth using the SPRING system.

Materials:
16-inch diameter pot
Soil
Seeds (e.g., corn)
Bottle to water
SPRING Soil Moisture Sensor/Manuel Sensor
SPRING Gateway
Data Collecting Sheet
Planting Diagram
YouTube Videos from Dashboard-Tutorial

Vocabulary:
Germination: When a seed begins to grow into a plant
Seed: A small object that can grow into a plant
Soil: Material that plants grow in
Moisture: Water in the soil
Data: Information we collect to track changes
Sensor: A tool that measures things like moisture and temperature

Engage
Entering the classroom carrying an empty pot and other planting materials. Asking students:
- What do you think we're going to do today?
- Do you remember what we learned yesterday?

Showing the materials and allow students to make predictions. If they mention the sensor or gateway, remind them that they will learn in the next lesson.

Explore
Planting a corn seed together as a class in one pot.
Guiding the process step-by-step while watching a video from the YouTube Dashboard or Classroom Setup playlist (optional).
Asking discussion questions during the activity:
- Why do we need soil? Why can't a seed just grow in water?

Encouraging students to think about what plants really need as they participate.

Explain
Introducing today's vocabulary one by one while pointing to real examples in the planting setup.
Writing short definitions on the board and placing labeled word cards around the pot.
Briefly discussing what each word means and how it connects to plant growth.

Elaborate
Dividing students into 3 data collection groups:
- Morning, Noon, and Afternoon / or hourly

Each group will check the plant during their time slot and observe daily changes.

Evaluate
Distributing the "Planting Diagram" activity sheet. Having students label parts of the planting process, use vocabulary in context, and draw what they think is happening under the soil as the seed begins to grow. (The picture in the diagram provided as a baseline.)

Notes

Date (MM/DD/YYYY)	Weather Today	Classroom Settings	Soil Moisture Sensor Data

Planting Diagram

[14]

SEED to SPRING Lesson 3

Subject: Middle School Agriculture	**Standards:** EA-8.3 Explore modern technology applications within the agriculture industry
Topic: Modern Tech Apps in Agriculture	**Lesson Objective:** By the end of the lesson, students will be able to: ●Identify IoT components: sensors and gateways ●Describe LoRaWAN and its key characteristics, specifically range and power ●Explain real-world use cases and benefits of IoT in agriculture ●Recognize related cloud services such as network server, database and dashboard
Materials: Soil Moisture Sensor SPRING Gateway "Soil Talking to the Farmer" image Crossword Puzzle Activity (online) LoRaWAN/Cloud Explainer Video (linked via dashboard or YouTube)	**Vocabulary:** Sensor: A tool that measures things like moisture or temperature Gateway: A device that sends sensor data to the internet LoRaWAN: A wireless network that works over long distances with little power Cloud: Online computers that help store, process, and display sensor data Dashboard: A visual display of the data to help make decisions Irrigation: The process of watering crops

Engage

1.Activity: Unhealthy Soil

a. "Imagine you are a farmer and notice that your plants do not look healthy. You suspect the problem might be the soil, but you are not sure whether it is too dry or wet. Here comes the question: How can we tell if the soil is wet or dry without digging it up or getting our hands dirty?"

i. Turn it into a fun guessing game. Let students shout out ideas.

2.Activity: Soil Talking to Farmer

a. Show the image "Soil Talking to the Farmer" to students and ask them to guess what each part does.

i. "The image below illustrates one way the soil can "talk" to the farmer. Can you tell what's happening?"

ii. Clarify any misconceptions right away.

Explore

3. Activity: Explore in Hand

a. Break into groups and explore the sensors and gateways that the teacher will hand out.

i. SE01-LB: Soil Moisture & Temperature Sensor

ii. RAK RPi Kit or SenseCap Indoor Gateway

4. Sensor Types, Use Cases and Benefits

a. Three different sensors

i. Soil moisture sensor: Measures how much water is present in the soil

ii. Temperature sensor: Measures the temperature of the air or the soil

iii. Rain gauge: Measures how much rainfall has occurred in a specific area

b. Show images of these sensors

c. Benefits

i. Irrigation management (Simplified as "How much to water crops")

ii. Tracking dry conditions

iii. Monitoring soil and plant health

d. Activity: Brainstorming
    i. Break into groups and brainstorm
        1. "What problems on a farm could sensors solve?"
        2. "Imagine you are designing an IoT solution for a farm, what sensors would you use?"

Explain

1.What is Internet of Things (IoT)?
a. IoT definition
b. Real-life examples
    i. Smartwatch sending your heart rate to your phone
    ii. Voice assistant playing your favorite music
c. Key components:
    i. Sensor: Measures specific physical properties in the environment through its probes
        1. Agricultural example: Soil moisture sensor measures how wet the soil is.
    ii. Gateway: Collects data from sensors, and sends them to the internet
2. Why Gateways Matter: A Look into LoRaWAN
a. Watch the video to learn about LoRaWAN and its characteristics
    i. Exemplified over the previous image "Soil Talking to the Farmer"
b. Key Points:
    i. Coverage: Connecting many sensors
    ii. Power: Low power consumption
    iii. Range: Long range transmission

Elaborate

a. The students are allowed to invent their own imaginary sensors just for fun
5. What is Cloud?
a. Cloud: It is made up of powerful computers that store, process, and visualize data online.
b. Watch the video about cloud components
c. Make a short summary of the components
    i. Network server, accepting and organizing the data
    ii. Database, storing the organized data
    iii. Dashboard, turning the stored data into visualizations that help farmers make smart decisions.

Evaluate

1. Play the Crossword Puzzle
    a. The students can play individually or in pairs/groups by creating an invite code inside the puzzle.

Notes

SEED to SPRING Lesson 4	
Subject: Middle School Agriculture	**Standards:** EA-8.3 Explore modern technology applications within the agriculture industry
Topic: Exploring the Dashboard	**Lesson Objective:** Students will explore the SPRING dashboard and weather station data, understand how farmers use these tools to make decisions, and identify the types of insights that can be gained from real-time agricultural data.
Materials: SPRING dashboard "SPRING Detective" activity sheet	**Vocabulary:** Weather Station: A tool that collects local weather information (e.g., temperature, humidity, rainfall) Trend: A pattern in data over time Variable: Something that can change and be measured (e.g., soil moisture, air temperature) Data Interpretation: Understanding what numbers and graphs mean in context

Engage

Entering the classroom with a "mystery" printed chart from the dashboard (no labels, just numbers/graph). Asking:

- What do you think this information is showing?
- If you were a farmer, why might you care about this information?

Letting students guess before revealing that it's from the SPRING dashboard and a nearby weather station.

Explore

Opening the actual dashboard on the projector.

- Show soil moisture, temperature, and weather station readings.
- Allow students to click around (or guide clicks) to discover different data views.

Giving them the SPRING Detective sheet and ask them to:

- Record three interesting observations from today's data.
- Write one farming decision they might make using this data (watering, harvesting, protecting plants, etc.).

Explain

Briefly introduce each vocabulary term.
Show where the "variables" appear on the dashboard.
Explain that farmers use both crop data (like soil moisture) and weather data to plan actions.

Elaborate

Grouping students in pairs:

- Giving each group one "farmer scenario" (e.g., Soil moisture dropping for 3 days + no rain).
- Asking them to decide what action the farmer should take and explain their reasoning using the data.

Evaluate

Groups share their decisions with the class and discuss the feasibility of their plans.

Notes

SPRING Detective Sheet

Name:_____ Our Class SPRING Location:

Date: __/__/____ _____

Today you're a SPRING Detective! Use the dashboard to find answers to these questions. Write clearly so other detectives can check your work.

Section 1 – Current Clues (today's readings)

1. What is the current temperature? _____ °C / °F
2. What is the current soil moisture? _____ %
3. What is the current humidity? _____ %
4. Is rain in the forecast for today? Circle: YES / NO

Section 2 – Data Trends (past readings)

5. In the last 3 days, has soil moisture gone up, down, or stayed the same?

_____

6. Over the past week, what was the highest temperature? _____ °; lowest temperature? _____ °

7. Look at the temperature graph — is it steady, rising, or falling?

_____

8. Find the day with the lowest soil moisture. What happened in the weather that day? _____

Section 3 – Detective Decisions (apply what you learned)

9. If soil moisture is low and no rain is coming, what should the farmer do?

_____

10. If heavy rain is coming tomorrow, what should the farmer prepare for?

_____

11. Which variable (soil moisture, temperature, rainfall) do you think is most important to check today? Why? _____

12. Imagine you are the farmer — what is one action you would take today based on the data? _____

Farmer Scenario Cards

Scenario 1 – Dry Days Ahead

Soil moisture has dropped from 30% to 20% in the last 3 days. The weather station shows no rain forecast for the next 4 days.

Question for your group: What should the farmer do? Why?

Scenario 2 – Too Much of a Good Thing

The last 2 days had heavy rainfall. Soil moisture is now 95%, and the temperature is cool.

Question for your group: What steps should the farmer take to protect the plants?

Scenario 3 – Perfect Timing?

The crop is almost ready to harvest. The forecast shows heavy rain coming tomorrow.

Question for your group: Should the farmer harvest now or wait? Why?

Scenario 4 – Sudden Heat Wave

Temperature jumped from 22°C to 35°C in one day. Soil moisture has started dropping quickly.

Question for your group: What should the farmer do to protect the plants?

Scenario 5 – (can be added more by teacher)

SEED to SPRING Lesson 5	
Subject: Middle School Agriculture	**Standards:** EA-8.3 Explore modern technology applications within the agriculture industry EA-3.4 Identify basic soil properties
Topic: Farm Data Comparison	**Lesson Objective:** Students will compare agricultural conditions from two different farms using both real-world news events and dashboard data. They will identify how environmental factors can lead to different farming decisions and outcomes.
Materials: Two short news articles -New York -Purdue corn field SPRING dashboard	**Vocabulary:** Impact: The effect of an event or condition on crops. Comparison: Looking at similarities and differences. Environmental Factor: A condition in nature (e.g., temperature, rainfall, frost) that affects plant growth. Decision-Making: Choosing an action based on information or data.

Engage

Entering the classroom with two printed news: "Late Spring Frost Damages Apple Orchards in New York" and "Purdue Corn Fields Show Steady Growth This Season"
Holding them up and asking:

- What differences do you think might explain these headlines?
- How might the farmers in each location feel?

Letting students share quick predictions

Explore

Reading a short excerpt from the articles.
Students will discuss with their pairs: "What kind of data might confirm or contradict what these articles says?"
Opening SPRING dashboard and briefly showing Purdue corn field data and New York apple achard data from the same week.
Students record 2-3 key differences/similarities in conditions between the two sites

Explain

Reviewing vocabulary terms and point out where they connect to the article and the data shown
Highlighting that real-world decision-making often comes from both data and field observation/news.

Elaborate

In small groups, students brainstorm: "If you were the scientist, how would you warn the farmers?" "If you were the farmers, what kind of decisions will you take?"
Groups also consider: "How would your plan change according to the place change?"
Encouraging students to use both the article and dashboard data to explain their reasoning.

Evaluate

Each group presents their plan shortly to the class.
The class votes on which plan seems most realistic given the data and the article.

Notes

News links:

- https://newyorkagconnection.com/news/new-york-apple-orchards-feel-the-effects-of-may-frost?
- https://www.brownfieldagnews.com/news/dry-conditions-impacting-corn-as-pollination-approaches

SEED to SPRING Lesson 6	
Subject: Middle School Agriculture	**Standards:** EA-3.4 Identify basic soil properties
Topic: Soil Properties	**Lesson Objective:** Students will observe and test different soil types to identify their properties, such as texture, drainage, and water-holding capacity, and connect these findings to how the SPRING system's sensors measure and monitor soil conditions in real time.
Materials: Different soil samples (sandy, clay, loam) Small clear cups Water droppers Magnifying glasses Paper towels, printer paper, and cardboard Soil properties observation chart SPRING dashboard	**Vocabulary:** Absorption: how much water a material can take in Drainage: how quickly water moves through a material Texture: how soil feels (sand, silt, clay mix) Retention: how well soil holds water Nutrients: elements in soil that plants need to grow

Engage

Holding up the three paper types (paper towel, printer paper, cardboard). Asking:

- If we spilled water, which one would soak it up fastest?
- Which one would hold the water the longest?

Letting students make predictions.

Quick demo: Pouring the same amount of water to each paper type and observing the speed and how much water is held. Asking:

- What does this make you think about when it comes to different soils?

Briefly explain: "Just like paper absorbs and holds water differently, soils do too — and our SPRING sensors measure these differences to help farmers know when and how much to water."

Explore

Setting up "Soil Investigation Stations" similar to the paper test.

1. Feel Test – rub a pinch of soil between fingers, note if it feels gritty, sticky, or smooth.
2. Water Test – pour a set amount of water in to soil and time drainage.
3. Visual Check – note color, visible particles, and smell.

Students rotate through each station, recording results on the observation chart.

Before moving to Explain, show a live snapshot from the SPRING dashboard of your classroom plant's soil moisture reading. Ask students to predict which soil type from today's test might behave most like the reading they see.

Explain

Introducing the vocabulary terms, linking them to what students saw in both the paper demo and the soil tests.

Elaborate

Grouping students and letting them choose a soil sample and discuss:

- Which crop might do well here?
- What problem could happen if it rained too much or too little?
- How could a SPRING sensor help the farmer using this soil?

Evaluate

Groups share their ideas with the class.

Tying back to how farmers need to know soil properties and monitor them with technology before planting.

Preview Day 7:

"Tomorrow, we'll see how our classroom soil compares to farms in different locations and use a Soil Texture Triangle to classify them."

Notes

Soil Properties Observation Chart

Name:_____ Date:__/__/____

Soil Sample Type	Texture (gritty, sticky, smooth)	Color	Smell (earthy, no smell, other)	Drainage Time (sec)	Water Retention (holds water well, drains fast)	Notes (any other observations)
Sandy Soil						
Clay Soil						
Loam Soil						
Other (if used)						

SEED to SPRING Lesson 7	
Subject: Middle School Agriculture	**Standards:** EA-3.4 Identify basic soil properties EA-8.3 Explore modern technology applications within the agriculture industry
Topic Soil Layers	**Lesson Objective:** Students will use their Day 6 soil test results, the Soil Texture Triangle, and SPRING dashboard data to classify soil types and understand how soil conditions affect crop decisions in real-world farming
Materials: Soil Texture Triangle Day 6 Soil Properties Observation Charts (completed by students) Teacher-provided sand/silt/clay percentage estimates for each sample SPRING dashboard (optional: to show soil moisture trends for local site)	**Vocabulary:** Absorption: how much water a material can take in Drainage: how quickly water moves through a material Texture: how soil feels (sand, silt, clay mix) Retention: how well soil holds water Nutrients: elements in soil that plants need to grow Soil Texture Triangle: a chart used to classify soil based on the percentage of sand, silt, and clay Loam: a balanced soil type with good drainage and nutrient retention Soil Type: the category a soil falls into based on texture and properties (e.g., sandy loam) Soil Profile: layers and characteristics of soil in a location

Engage

Asking quick "what-if" scenarios:

- If you tried to plant tomatoes in the dessert, what would happen?
- What if planted rice in sandy soil with no irrigation?
- What would happen to carrots if the soil stayed muddy all months?

Students share predictions. Connecting answers to soil properties as the key factor for crop success.

Explore

Reviewing Day 6 soil test results as a class.
Providing sand/silt/clay percentage estimates for each sample (based on Day 6 observations).
Showing the Soil Texture Triangle and explain how farmers use it to classify soil types.
Displaying SPRING dashboard soil moisture data for the classroom plant and Purdue farm for the past week.
Students compare the dashboard readings to their soil test results, noting how soil type influences water retention.

Explain

Explaining to students how to read the Soil Texture Triangle.
Classifying one sample together as a demonstration.
Relating each vocabulary term to the Day 6 test results and dashboards readings.

Elaborate

Group activity:
Students classify their assigned soil sample using the Soil Texture Triangle.
Groups decide:

- Which crops would grow best in that soil
- What challenges could occur (too much/little rain, irrigation needs)
- Groups compare their answers with dashboard readings from Purdue farm to discuss similarities and differences.

Evaluate

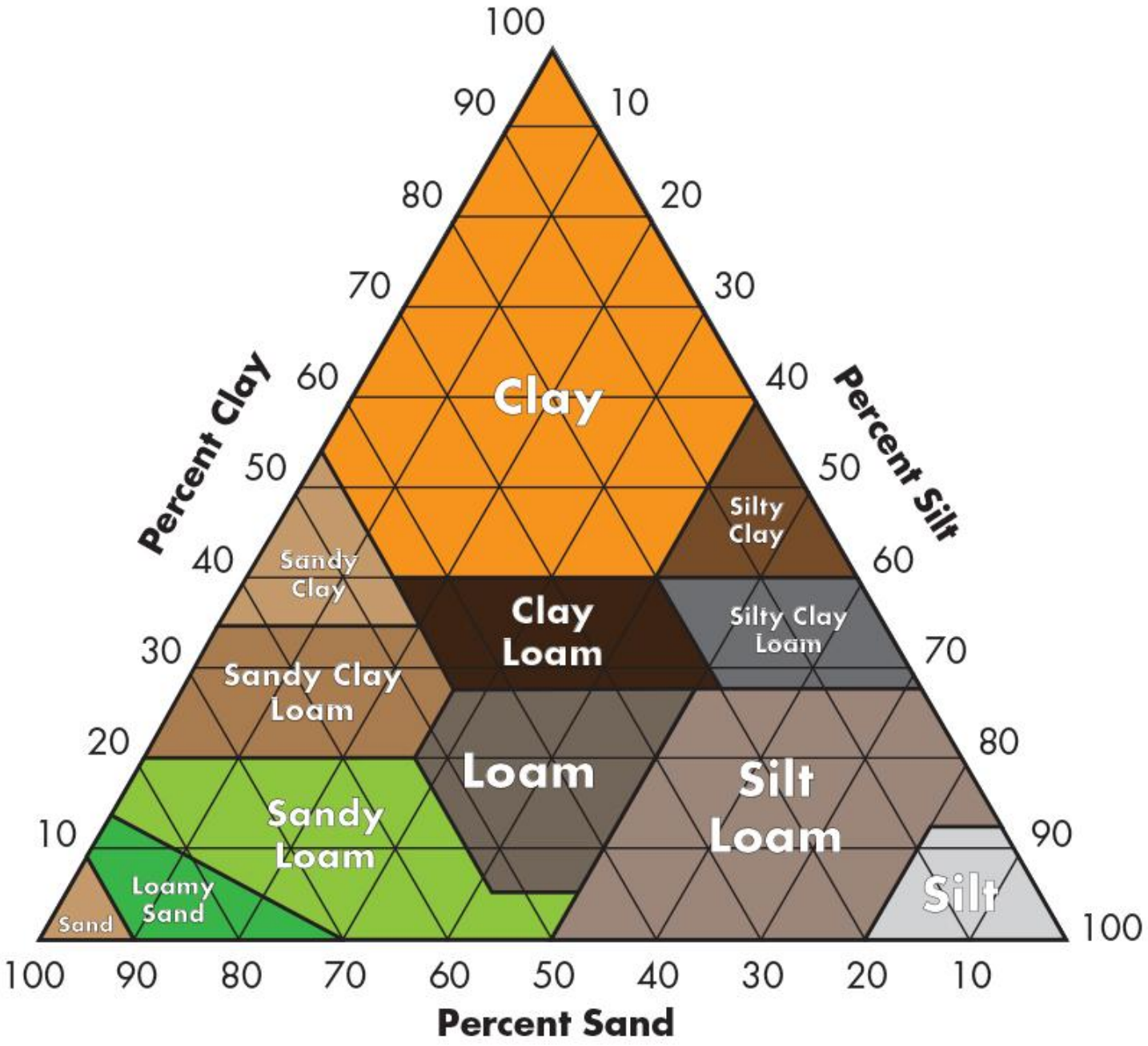
Each group shares their soil classification and farming recommendations.

Class discusses which recommendations are most realistic based on both soil test results and real sensor data.

Teacher emphasizes that farmers combine field testing and real-time technology like SPRING sensors to make informed crop decisions.

Notes

Soil Properties Triangle



SEED to SPRING Lesson 8	
Subject: Middle School Agriculture	**Standards:** EA-3.1 – Understand basic needs of plant growth from germination to soil fertility EA-3.4 Identify basic soil properties EA-8.3 Explore modern technology applications within the agriculture industry
Topic: What We Learned?	**Lesson Objective:** Students will synthesize what they have learned about soil properties, plant growth needs, and agricultural technology by using SPRING dashboard data and classroom experiment results to explain how soil and environmental conditions affect crop success.
Materials: Students' notes from Days 1 to 7 SPRING dashboard Photos of the class plant from germination to current state "Plant Growth & Soil Connection" worksheet	**Vocabulary:** Germination: when a seed begins to grow into a plant Fertility: the ability of soil to provide nutrients needed for plants to grow Absorption: how much water a material can take in Drainage: how quickly water moves through a material Retention: how well soil holds water Nutrients: elements in soil that plants need to grow Soil Texture Triangle: a chart used to classify soil based on sand, silt, and clay percentages Agricultural Technology: tools and systems used to improve farming efficiency (e.g., sensors, dashboards, automated irrigation)

Engage

Showing the class two photos side-by-side:

- The class corn plant on Day 2 (just planted)
- The current plant on Day 8

Asking:

- What do you think helped our plant grow this far?
- What challenges did it face?

Writing student ideas on the board and circle anything related to soil or technology.

Explore

Opening the SPRING dashboard and reviewing key data trends from the class plant.
In pairs, students compare the dashboard trends with their Day 6 soil test results and Day 7 soil classification.
Students fill in the "Plant Growth & Soil Connection" worksheet with:

- Soil property
- Effect on plant growth
- Evidence from dashboard data or observation

Explain

Reviewing the plant growth stages (germination -> seedling -> mature plant)
Linking each stage to what plants need most (water, nutrients, light, temperature) and how soil fertility plays a role.
Connecting this to the SPRING project:

- Sensors: collect continuous soil & environmental data
- Dashboard: helps farmers monitor and respond to plant needs in real time
- Decision-making: changing irrigation or fertilization based on data

Elaborate

Small-group challenge:

- Scenario: "You are hired to advise a farmer who wants to grow the same crop as our classroom plant but in a different location."

Groups decide:

- What soil properties the farmer should look for
- What technology tools (like SPRING sensors) would be most helpful
- How they would monitor and adjust plant care over time

Groups prepare a short action plan (bullet points) using both their soil learning and dashboard experience.

Evaluate

Each group shares one recommendation from their action plan.

Teacher checks that students are making clear connections between soil properties, plant needs, and technology.

Class discusses: "How would farming be different without technology like SPRING?"

Notes

Plant Growth & Soil Connection Worksheet

Name:_____ Date:__/__/_____

Instructions:
Use your Day 6 soil test results, Day 7 soil classification, and SPRING dashboard data to complete this chart. For each soil property, explain how it affects plant growth and give evidence from your observations or data.

Soil Property	Effect on Plant Growth (Why it matters)	Evidence from Experiment or Dashboard
Absorption (how much water soil can take in)		
Drainage (how quickly water moves through)		
Retention (how well soil holds water)		
Texture (sand, slit, clay, mix)		
Nutrients (elements plants need to grow)		
Soil type (e.g., sandy loam, clay)		
Soil Fertility (ability to support plant growth)		

Based on your findings, what changes would you recommend improving soil conditions for our plant?

SEED to SPRING Lesson 9	
Subject: Middle School Agriculture	**Standards:** EA-9.2 Describe career opportunities and means to achieve those opportunities in agriculture EA-8.3 – Explore modern technology applications within the agriculture industry
Topic: Field Trip	**Lesson Objective:** Students will consolidate their understanding of soil, plant growth, and agricultural technology by engaging directly with a farmer or agricultural expert who uses tools similar to the SPRING system. They will connect their classroom learning to real-world practices through guided observation and a structured Q&A session.
Materials: Interview Questions Worksheet	**Vocabulary:** Career Pathway: A planned route to gain skills and qualifications for a specific job. Agronomist: A scientist who studies crops and soils. Agricultural Engineer: Designs systems and technology to improve farming. Extension Agent: An educator who helps farmers apply research in the field. Precision Agriculture: Using technology and data to improve farming decisions.

Engage

Teacher leads a brief recap of what students have learned in the SPRING project: planting, soil testing, moisture monitoring, and decision-making based on data.
Pose the question: "How do farmers use these same ideas and tools in the real world?"
Students review the Interview Questions Worksheet and choose at least 3 questions they are most interested in asking.

Explore

Students visit the farm or agricultural site and have time to observe the surroundings, crops, and any visible technology in use (e.g., sensors, irrigation systems, or dashboard data).

Explain

The farmer/expert describes the farm's conditions, crops, and the role of technology in daily decision-making, making explicit connections to the types of data and tools used in the SPRING project.

Elaborate

Students use their Interview Questions Worksheet to guide a Q&A with the expert/farmer.

Evaluate

At the end of the visit, students complete the reflection section on their worksheet, summarizing:

- One new thing they learned about agricultural careers.
- One example of how technology improves farming.
- A connection between today's visit and the SPRING project work in class.

Notes

Farm Address: Purdue Research Farm - High Bay, 4530 US-52, West Lafayette, IN 47906
Primary Contact Person: Dr. Yaguang Zhang, zhan1472@purdue.edu
Department Contact: info@oatscenter.org

SEED to SPRING – Field Trip Interview Questions Worksheet

Name:_____ Date:__/__/_____

Expert/Farmer's Name: _____

Part 1 – Before the Interview
Choose at least 3 questions below to ask during your interview. You can add your own at the end.

1. What crops do you grow here, and why did you choose them?
2. How do you decide when and how much to water your crops?
3. Have you ever tested your soil? If yes, what did you learn?
4. What kind of technology do you use on your farm (e.g., sensors, drones, dashboard systems)?
5. How does weather data affect your decisions?
6. Have you ever had to change crops or planting methods because of soil conditions?
7. What is one challenge you face in farming that technology helps solve?
8. How is your work similar to what we did in the SPRING project at school?
9. What advice would you give someone interested in a career in agriculture?
10. (Your own question)
    _____

Part 2 – Interview Notes
Write short notes or key phrases during the interview:

SEED to SPRING Lesson 10	
Subject: Middle School Agriculture	**Standards:** EA-3.1 – Understand basic needs of plant growth from germination to soil fertility EA-3.4 Identify basic soil properties EA-8.3 Explore modern technology applications within the agriculture industry
Topic: Assessment & Reflection	**Lesson Objective:** Students will demonstrate their understanding of plant growth needs, soil properties, and modern agricultural technology by completing a final project and participating in a review activity. They will also reflect on how the SPRING project has changed their view of farming and technology.
Materials: Final Quiz Student data and notes from previous lessons SPRING dashboard access	**Vocabulary:** Light, Water, Temperature, Nutrients, Absorption, Drainage, Retention, Texture, Soil Texture Triangle, Loam, Precision Agriculture, Sensor, Dashboard

1. Beginning with a brief introduction (2–3 minutes) reminding students this is their final check-in and a chance to show what they've learned.
2. Allowing 30 minutes for students to complete the quiz individually.
3. Using the remaining class time to:
    - Review quiz answers as a group.
    - Discuss key takeaways from the SPRING project (soil, plant growth, technology).,
    - Collect exit slips where students share one thing they learned and one question they still have.
4. Closing by thanking students and connecting their learning to real-world agriculture.

Notes

Answers of Final Quiz:

1. C) Plastic
2. A) Too hot or too cold can slow or stop germination
3. B) Roots struggle to grow due to poor drainage
4. B) Measure how much water is in the soil
5. C) Sandy
6. B) Nutrients
7. A) It prevents roots from rotting in too much water
8. C) Hot, dry weather
9. B) Cactus
10. B) Loam
11. C) Using technology and data to make farming decisions
12. A) Seeds may germinate more slowly
13. B) Lack of nutrients or water stress
14. B) Drains quickly
15. C) Extension agent

Final Quiz

Name:_____ Date:__/__/_____

1. Which of the following is NOT a basic need for plant growth?
A) Water
B) Light
C) Plastic
D) Nutrients

2. Seeds need the right temperature to germinate because:
A) Too hot or too cold can slow or stop germination
B) Heat makes seeds grow twice as fast
C) Cold always kills seeds instantly
D) Temperature has no effect

3. If your soil is 50% clay, 30% silt, and 20% sand, which problem is most likely?
A) Soil dries out too fast
B) Roots struggle to grow due to poor drainage
C) Soil blows away in the wind
D) Nutrients wash away quickly

4. In the SPRING project, the soil moisture sensor helps farmers:
A) Count how many seeds are in the soil
B) Measure how much water is in the soil
C) Predict the price of corn
D) Test soil color

5. Which soil type drains water the fastest?
A) Clay
B) Loam
C) Sandy
D) Silty

6. If plants in the SPRING project grew slowly even with enough water, what might be missing?
A) Soil
B) Nutrients
C) Sand
D) Shade

7. Why is good drainage important for healthy plant growth?
A) It prevents roots from rotting in too much water
B) It keeps the soil cold
C) It stops insects from entering the soil
D) It makes plants grow faster in all conditions

8. If the SPRING dashboard shows soil moisture dropping quickly for two days, what could be a possible cause?
A) Heavy rain
B) Overwatering
C) Hot, dry weather
D) Adding more clay to the soil

9. Which crop would likely do BEST in sandy soil with limited water?
A) Rice
B) Cactus
C) Lettuce
D) Carrots

10. Which soil type has a balanced mix of sand, silt, and clay?
A) Clay
B) Loam
C) Sandy
D) Silty

11. Precision agriculture means:
A) Guessing when to water crops
B) Planting in perfect rows only
C) Using technology and data to make farming decisions
D) Growing crops without soil

12. If soil temperature stays below the ideal range for two weeks, what might happen?
A) Seeds may germinate more slowly
B) Plants will grow twice as fast
C) Roots will dry out immediately
D) Soil will turn into clay

13. Yellowing leaves on a plant can be caused by:
A) Too much sunlight only
B) Lack of nutrients or water stress
C) Planting too close together only
D) Using a loam soil type

14. Which is an advantage of sandy soil?
A) Holds water well
B) Drains quickly
C) Contains more nutrients than clay
D) Always keeps plants warm

15. Which career works directly with farmers to apply research in the field?
A) Agronomist
B) Agricultural engineer
C) Extension agent
D) Soil tester

Code Your Answers Here!

1	A	B	C	D
2	A	B	C	D
3	A	B	C	D
4	A	B	C	D
5	A	B	C	D
6	A	B	C	D
7	A	B	C	D
8	A	B	C	D
9	A	B	C	D
10	A	B	C	D
11	A	B	C	D
12	A	B	C	D
13	A	B	C	D
14	A	B	C	D
15	A	B	C	D

Standards Covered:

- **EA-3.1**: Understand basic needs of plant growth
- **EA-3.4**: Identify basic soil properties
- **EA-8.3**: Explore modern technology applications within the agriculture industry

Criteria	Excellent (4)	Proficient (3)	Developing (2)	Beginning (1)
Understanding Plant Needs (EA-3.1)	Clearly explains all key plant needs (light, water, soil, temp) and how each affects growth using examples from the experiment	Explains most key plant needs with some connection to plant health	Identifies some plant needs but lacks clear explanation or connection	Struggles to identify basic plant needs or provides unrelated info
Soil Observation & Comparison (EA-3.4)	Accurately describes properties of each soil type (texture, moisture, color) and connects observations to plant growth	Describes most soil properties with some connection to plant health	Gives limited descriptions or weak connection to plant growth	Minimal effort or inaccurate observations of soil properties
Sensor Integration Understanding (EA-8.3)	Clearly explains how the SPRING system works (sensor, LoRaWAN, dashboard) and its role in the project	Demonstrates general understanding of the SPRING tech and purpose	Basic or partial understanding of how the system works	Unable to explain or recall how the SPRING system operates
Experiment Setup & Care (data collection)	Setup is complete, well-labeled, and consistently cared for over the 10 days	Setup is complete and maintained with minor issues	Incomplete setup or inconsistent maintenance	Setup missing, mislabeled, or not maintained
Final Presentation or Summary (as a group)	Effectively communicates findings, uses data, and reflects on what was learned	Communicates main points with some data or examples	Presentation lacks clarity or detail	Unclear or missing presentation component

25–28: Excellent – Strong understanding and clear reflection
21–24: Good – Mostly complete with small gaps
17–20: Basic – Some understanding, missing parts or confusion
17 -0: Needs Support – Key elements missing, limited effort

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Purpose and Outcomes

This 10-day module introduces middle school students to hands-on agriculture through classroom experiments using sensors and real-time data. Aligned with Indiana Agriculture standards, it fosters applied science, data literacy, and real-world problem solving through plant growth, soil study, and interactive dashboard.

Teachers Will:

- Lead a standards-based agriculture project with ready-to-use materials
- Integrate sensor tools and dashboards into classroom instruction
- Adapt the module to high-tech or low-tech settings

Students Will:

- Investigate plant needs and soil properties through direct experimentation
- Analyze real agricultural data using dashboard tools
- Explore careers and technologies in modern agriculture

Aligned with Indiana Academic Standards for Middle School Agriculture

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